Mawlana Bhashani Science and Technology University (MBSTU)
Department of ICT

University_Attendance

A Web-Based University Attendance Management System

Final Capstone Project, Part I: Proposal and Prototype

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1. Abstract

Most universities still record class attendance on paper sheets. Sheets get lost, totals are calculated by hand, students cannot check their own record, and corrections leave no trace. University_Attendance replaces the paper sheet with a centralized web application where teachers record attendance per class, students monitor their own percentage, and administrators manage courses and departmental reports. Every class and every correction is stored permanently.

2. Problem Statement and Objectives

Manual attendance causes slow reporting, calculation errors, proxy marking, no audit trail and no early warning for students at risk. The objectives are:

- Let teachers take attendance for a class in under one minute.
- Calculate attendance percentage automatically and correctly.
- Give each student a private view of their own record and warnings.
- Keep a permanent history of every class and every correction.
- Give administrators department-level reports.

3. Scope

In scope (version 1)	Out of scope (future work)
Teacher attendance taking, history, correction, report, CSV export; Student dashboard with percentage and low-attendance warning; Admin account and course management; Class duration of 1 to 5 hours; one Present or Absent mark per class counts for every hour of it; Course join link: students join with their university email and a nickname, ID and full name saved automatically; the teacher sees nickname and ID; Role-based access and audit log	Biometric or face recognition; Mobile native app; Exam eligibility decisions (the university policy decides, not the software); Integration with the university ERP

4. Stakeholders and Users

User	Main needs
Teacher	Fast attendance entry, history, safe correction, reports
Student	View own attendance, dates present or absent, warnings
Administrator	Manage users and courses, view department reports

5. Requirements

Functional

ID	Requirement
FR1	Every teacher, student and admin has an individual account (university email as login ID and a password) and, after signing in, sees only their own dashboard.
FR2	A teacher records Present or Absent for every enrolled student of their own course for a given date, one mark per student for the class.
FR3	The system rejects future dates and a second class on the same date for the same course.
FR4	A teacher can correct a record only with a reason; the change is stored in an audit log.
FR5	Attendance percentage = attended hours / held hours $\times$ 100, counting only held classes and only hours the student was enrolled for; shows N/A when none held.
FR6	A student sees per-course hour totals, a timeline of classes (date, hours, Present or Absent) that can be filtered to Present, Absent or All classes, and a warning below the configurable minimum (default 60%).
FR7	A teacher exports a course report as CSV.
FR8	An admin creates teacher and student accounts, courses, enrollments and assigns teachers.
FR9	A teacher generates a course join link (random code, expires in 7 days, can be replaced by a new one).
FR10	A student opens the link and signs in with the university email; the student chooses a nickname, the full name comes from the university account, the ID from the email (it24065@mbstu.ac.bd gives ID IT24065), and the student is enrolled automatically with no manual entry by the teacher.
FR11	The system rejects expired or invalid links, emails outside the university domain, emails missing from the university directory, duplicate joins and nicknames already used in the course; one university email maps to one student.

ID	Requirement
FR12	A teacher sets the class duration (1 to 5 hours) when taking attendance; each student gets one Present or Absent mark for the class, and it counts for every hour of that class (a 2-hour class gives 2 present hours or 2 absent hours).
FR13	The attendance sheet shows each student's nickname and ID; clicking the ID shows the full name, university email, nickname and attendance summary.
FR14	A nickname has 2 to 20 characters and is unique within a course.
FR15	Passwords are stored only as salted hashes in the database (users table). A password needs at least 8 characters with a letter and a number, and 5 wrong attempts lock the account for a minute.
FR16	Forgot password: the user enters the university email, receives a 6-digit reset code (valid 10 minutes, limited tries, single use) and sets a new password. The reply never reveals whether an email is registered.
FR17	Student IDs are always listed in order, newest batch first and then by number (IT24001, IT24002, …, IT23001, …), whatever order students join. The ID is added automatically from the university email, so nobody types it.
FR18	On a desktop browser the teacher can take attendance with the keyboard: W or Up arrow and S or Down arrow move the highlighted row, and Enter switches Present and Absent. The active row is highlighted automatically. Phones and tablets use touch.
FR19	The admin manages a university directory (email and full name). A student can join a course only if the email is in the directory, and the full name is taken from it. In the real system the directory is replaced by the university account (OAuth sign-in).

Non-functional

- **Security:** passwords stored only as salted hashes (bcrypt in the real system), account lockout after wrong attempts, password reset by a one-time emailed code, role-based authorization, students cannot edit records.
- **Reliability:** data stored in a relational database with regular backup.
- **Usability:** a 50-student sheet can be completed in under one minute; works on phone screens.
- **Performance:** a page loads in under 3 seconds for 500 concurrent users.

6. Technology Stack

Layer	Choice
Frontend	HTML, CSS, JavaScript
Backend	Java, Spring Boot, Spring Security (email and password login with bcrypt hashes; optional OAuth2 sign-in with the university account)
Database	MySQL with Spring Data JPA
Tools	Git and GitHub, JUnit, Mermaid and LaTeX for documents

7. Software Process Model: Incremental Model

The Incremental model fits this project because the core idea (take attendance, keep history, let students view it) can be delivered first and working, and later features are added in further increments. Each increment passes through requirements, design, implementation and testing, so risk is found early and the course deadline is protected.

Increment	Content	Output
1	Proposal, requirements, UML design	This document
2	Interface prototype (8 classes)	Working demo
3	Backend, database, authentication, roles	Working system
4	Testing, fixes, documentation, deployment	Released website

8. Schedule, Milestones and Deliverables

Milestone	Date	Deliverable
M1 Proposal approved	27 Oct 2026	Proposal, requirements, six UML diagrams, Gantt chart
M2 Prototype ready	27 Oct 2026	Interactive interface using 8 classes
M3 Backend and database done	29 Nov 2026	REST API, MySQL schema, login, roles
M4 Integration complete	6 Dec 2026	Frontend connected to backend
M5 Testing complete	13 Dec 2026	Test report, bug fixes
M6 Final release	20 Dec 2026	Deployed site, final report, source on GitHub

Gantt Chart

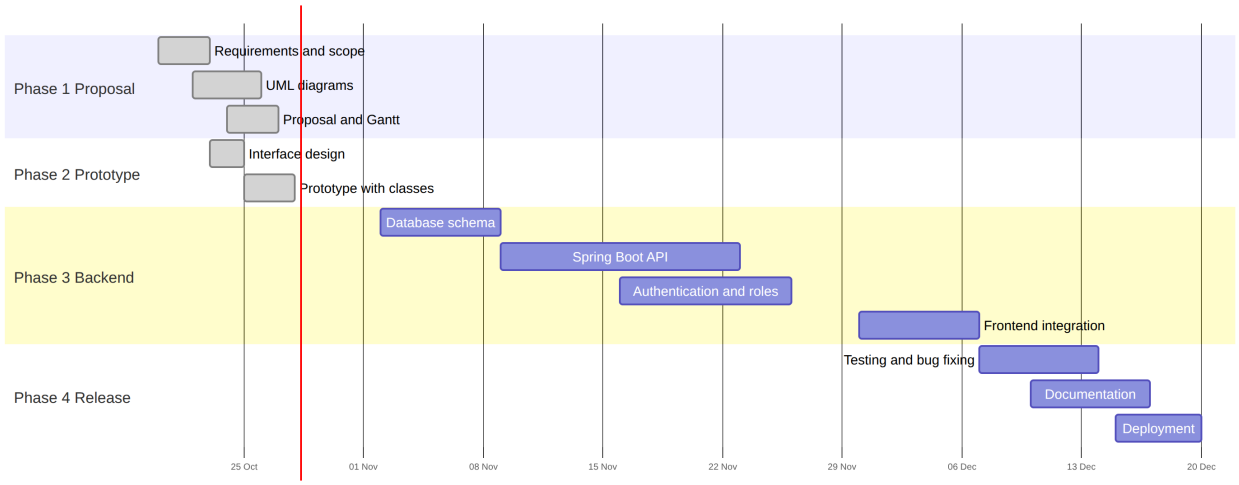


Figure 1: Project Gantt Chart

9. Risk Analysis

Risk	Likelihood	Mitigation
Learning Spring Boot takes longer	Medium	Prototype first; build backend in small increments
Scope grows	High	Version 1 limited to attendance, history, student view
Data loss	Low	Database backups, Git for code

Risk	Likelihood	Mitigation
Unauthorized edits	Medium	Role checks on the server, audit log

10. Testing Plan

Unit tests (JUnit) for percentage calculation, duplicate-date rule, future-date rule and permission checks. Integration tests for login and attendance save. User acceptance testing with one teacher and a few students. Example case: 20 attended hours out of 25 held hours must display 80%. The prototype also has automated rule tests (`frontend/tests/models.test.js`, 26 tests) for duration limits, percentage weighted by class hours, corrections, join rules, ID sorting, login, password rules and password reset.

11. UML Diagrams

11.1 Data Flow Diagram (Level 0)

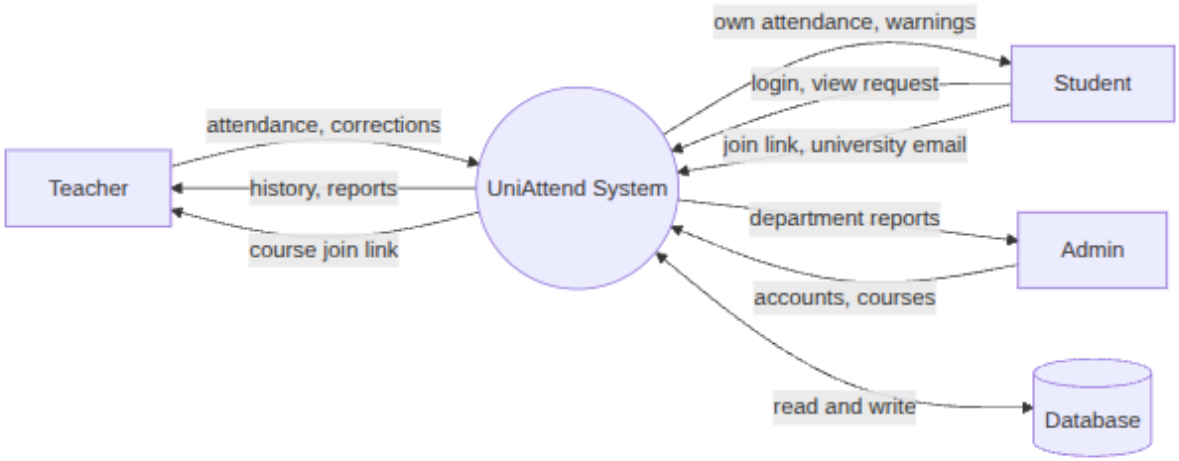


Figure 2: Data Flow Diagram (Level 0)

11.2 Use Case Diagram

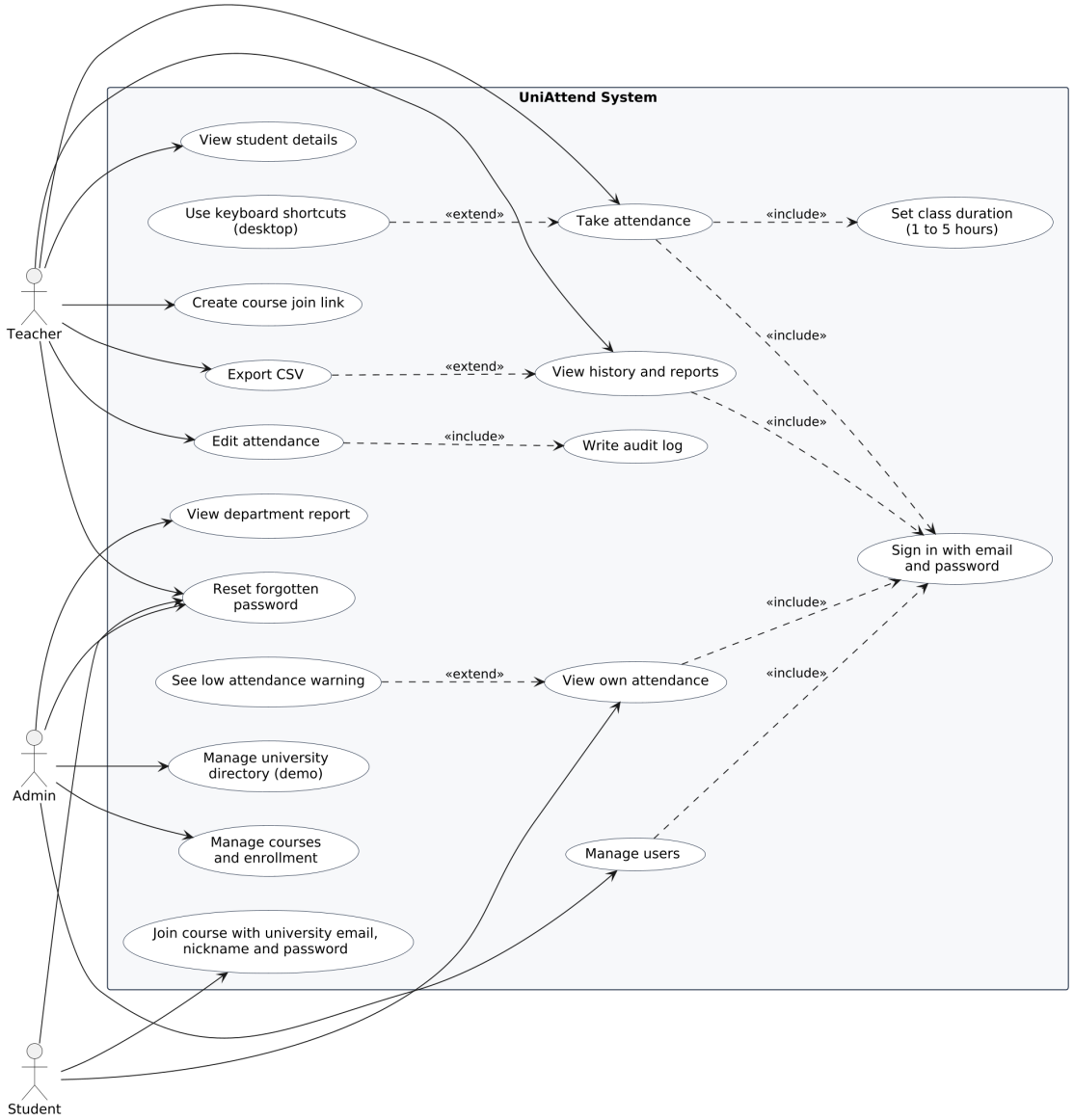


Figure 3: Use Case Diagram

11.3 Activity Diagram (Take Attendance)

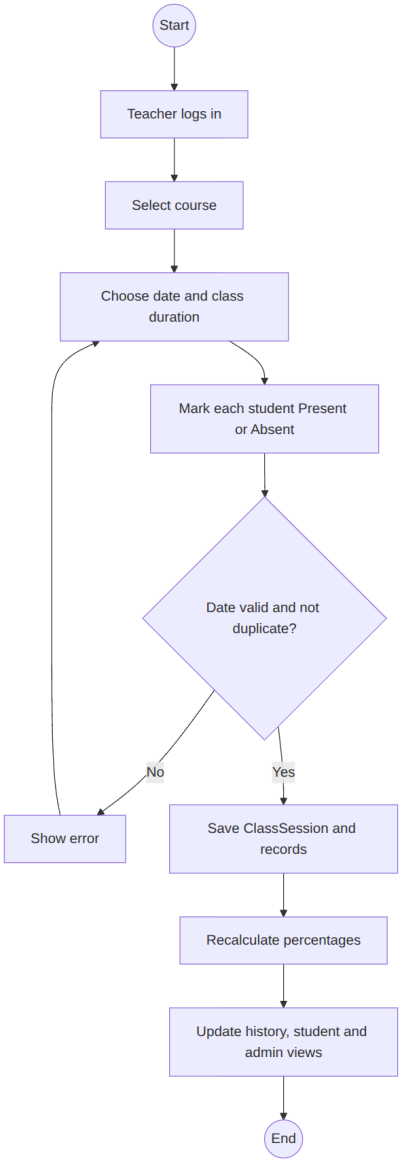


Figure 4: Activity Diagram: Take Attendance

11.4 Entity Relationship Diagram

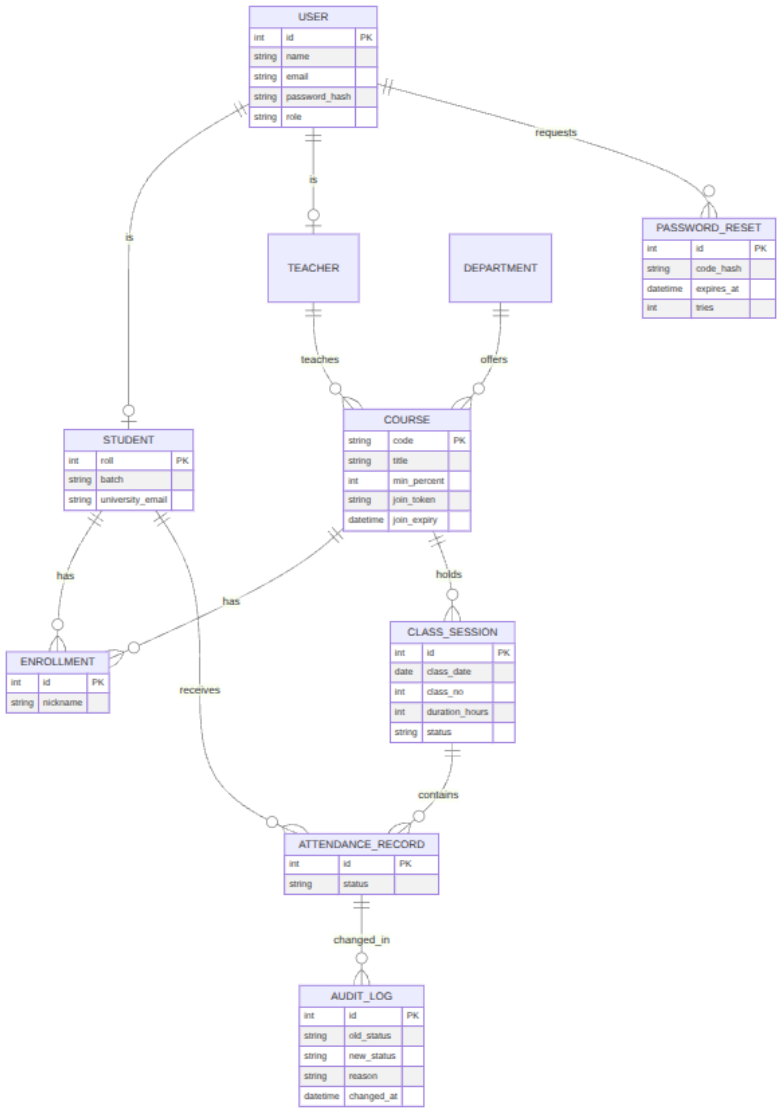


Figure 5: Entity Relationship Diagram

11.5 Class Diagram

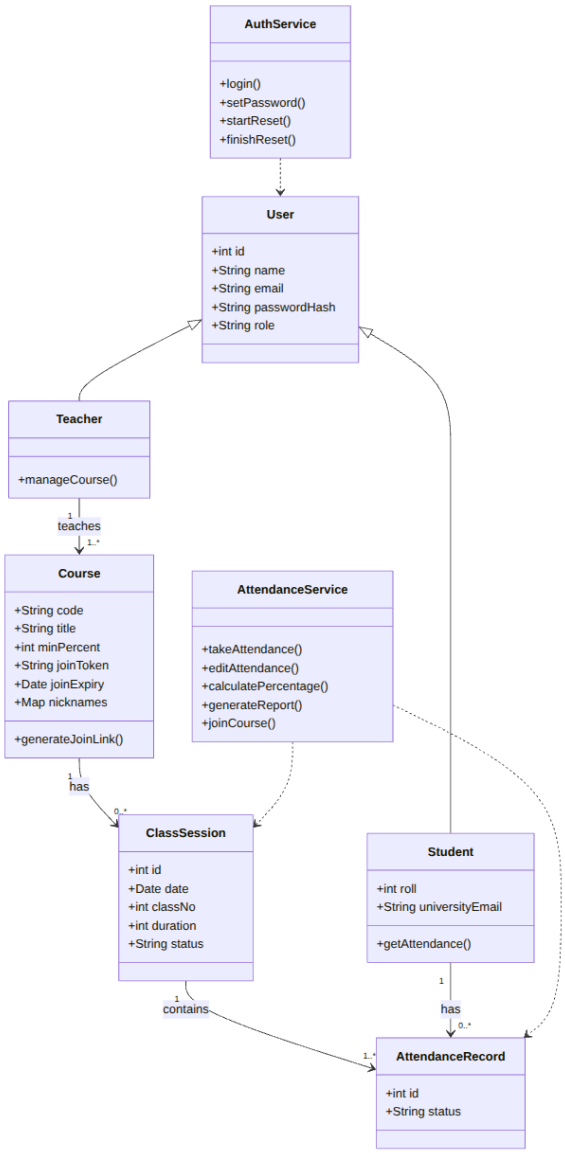


Figure 6: Class Diagram

11.6 Object Diagram

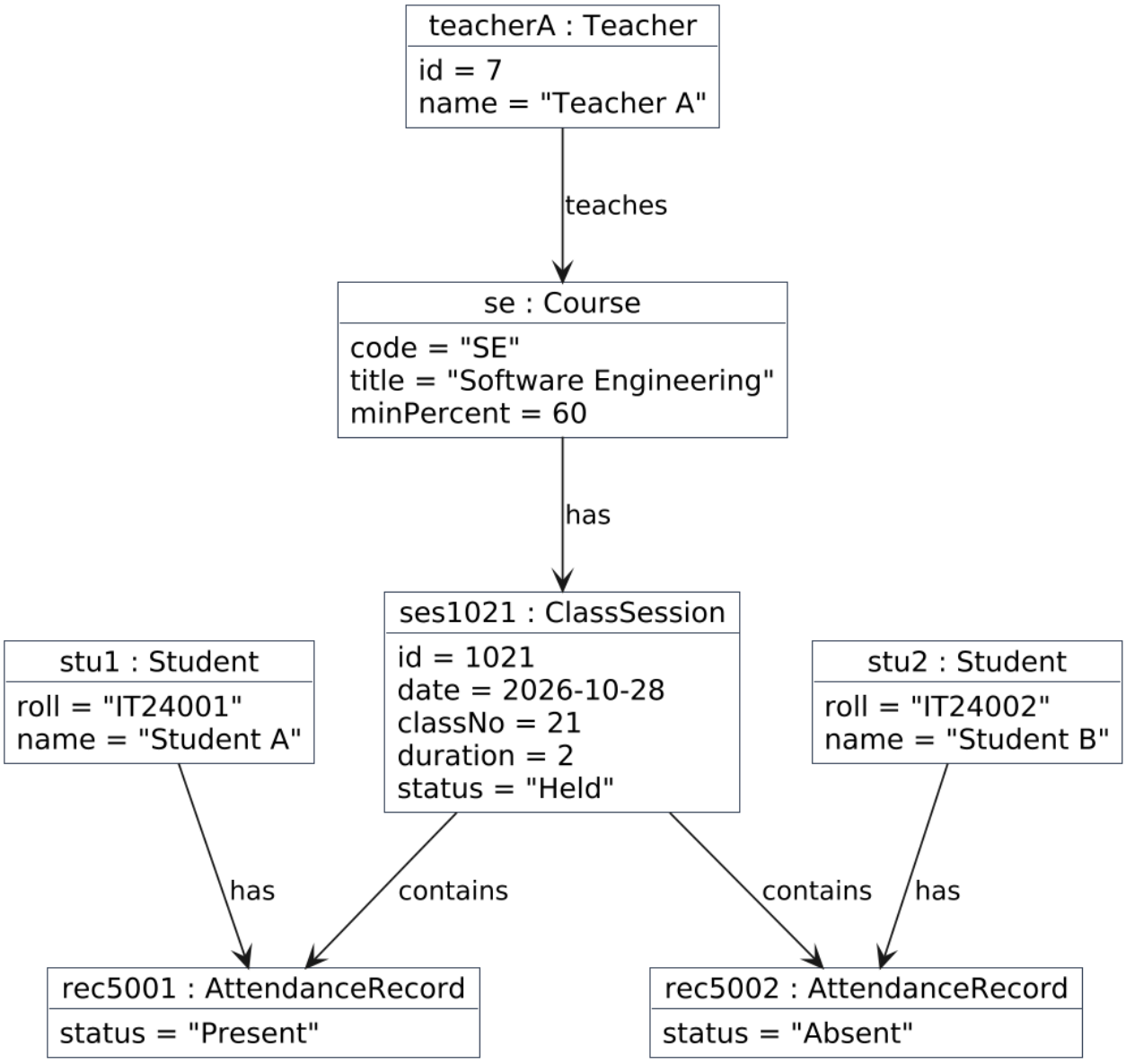


Figure 7: Object Diagram

11.7 Activity Diagram (Join Course by Link)

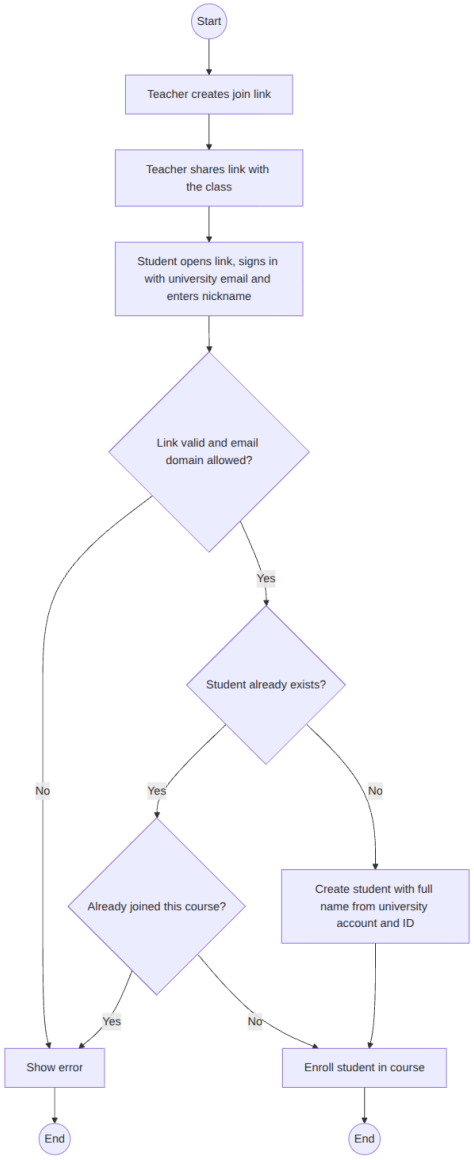


Figure 8: Activity Diagram: Join Course by Link

11.8 Data Flow Diagram (Level 1)

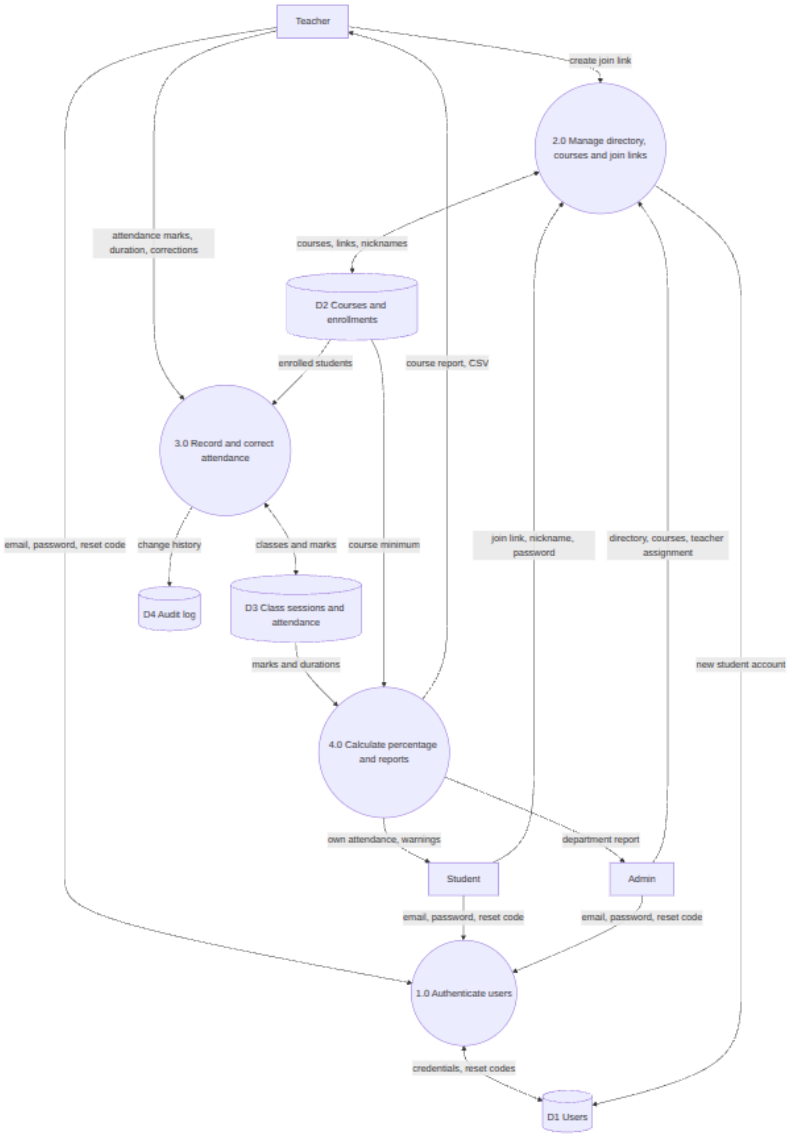


Figure 9: Data Flow Diagram (Level 1)

12. Interface Prototype

The prototype implements eight classes from the class diagram in JavaScript (with email and password login, forgot password, class duration of 1 to 5 hours counted as hours, keyboard attendance on desktop, sorted student IDs, a student timeline of hour squares, nicknames, and the full name collected automatically from the university account): `User`, `Student`, `Teacher`, `Course`, `ClassSession`, `AttendanceRecord`, `AttendanceService` and `AuthService`. It demonstrates taking attendance, history, correction with an audit log, the percentage rule, the student warning and the admin report. Open the prototype. Prototype data is saved in the browser's localStorage only (it works across tabs of one browser, and is not a database); the full system stores it in MySQL.

Prototype Screenshots

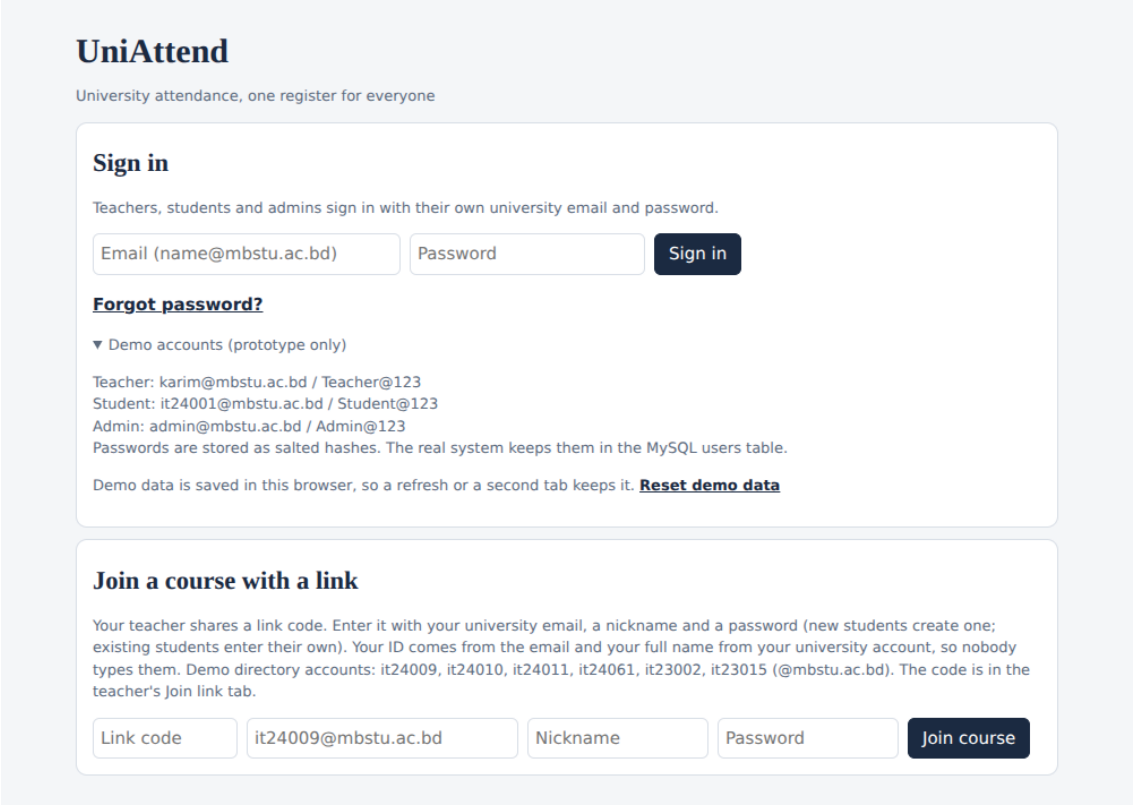


Figure 10: Sign in with university email and password (demo accounts, forgot password link, join by link)

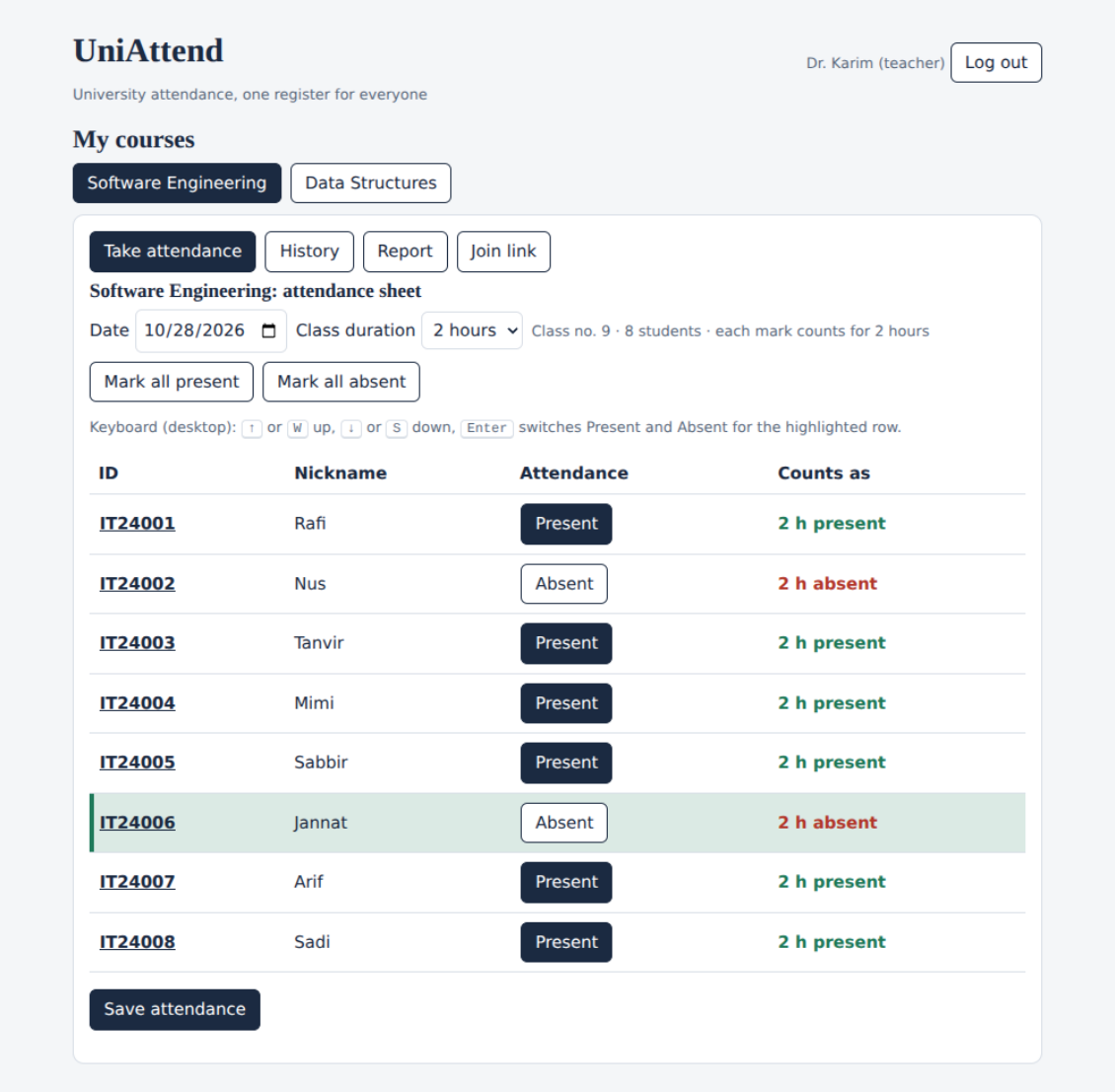


Figure 11: Teacher takes attendance with the keyboard: highlighted row, duration selector, nickname and ID, one mark per student

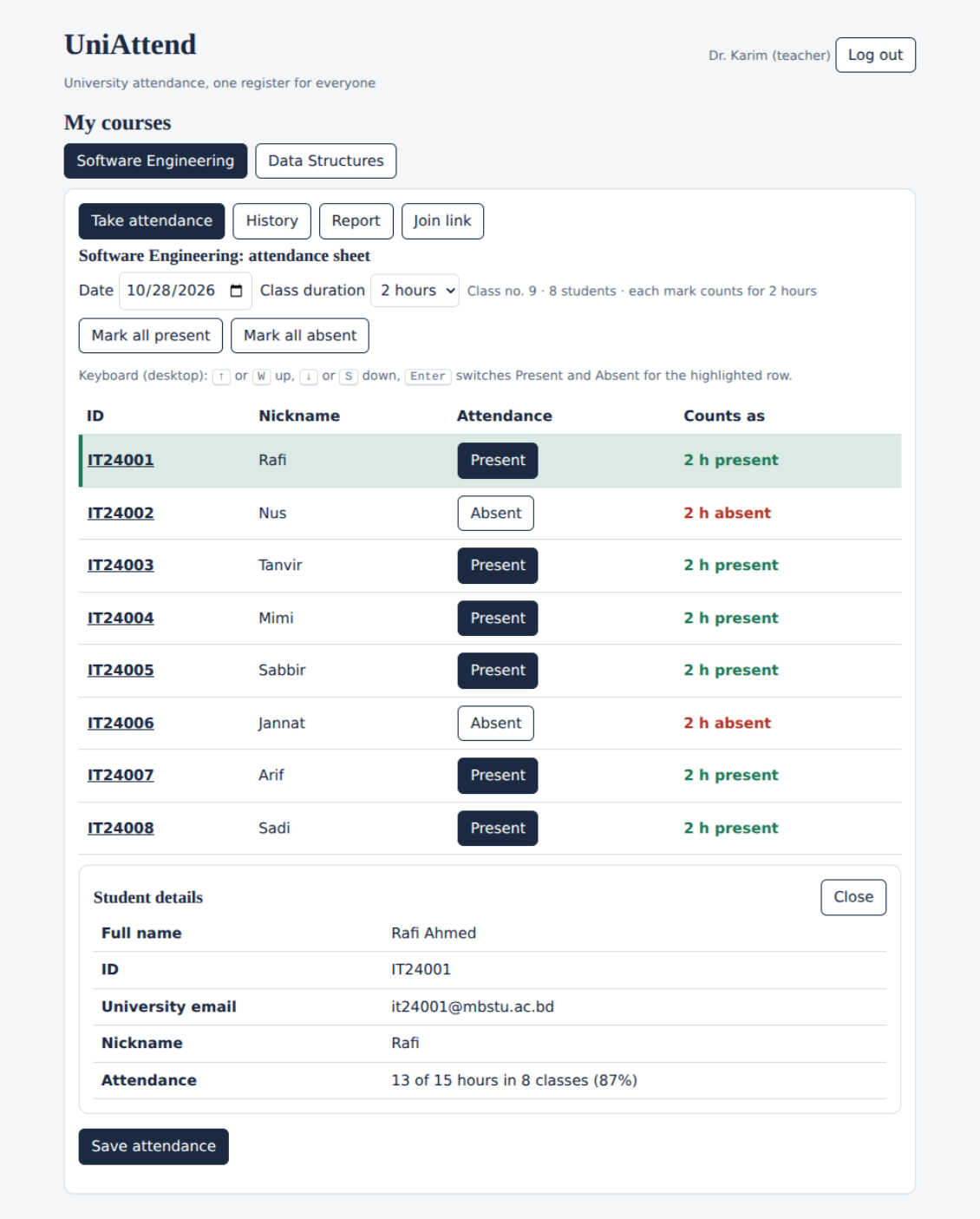


Figure 12: Click an ID to see full name and details

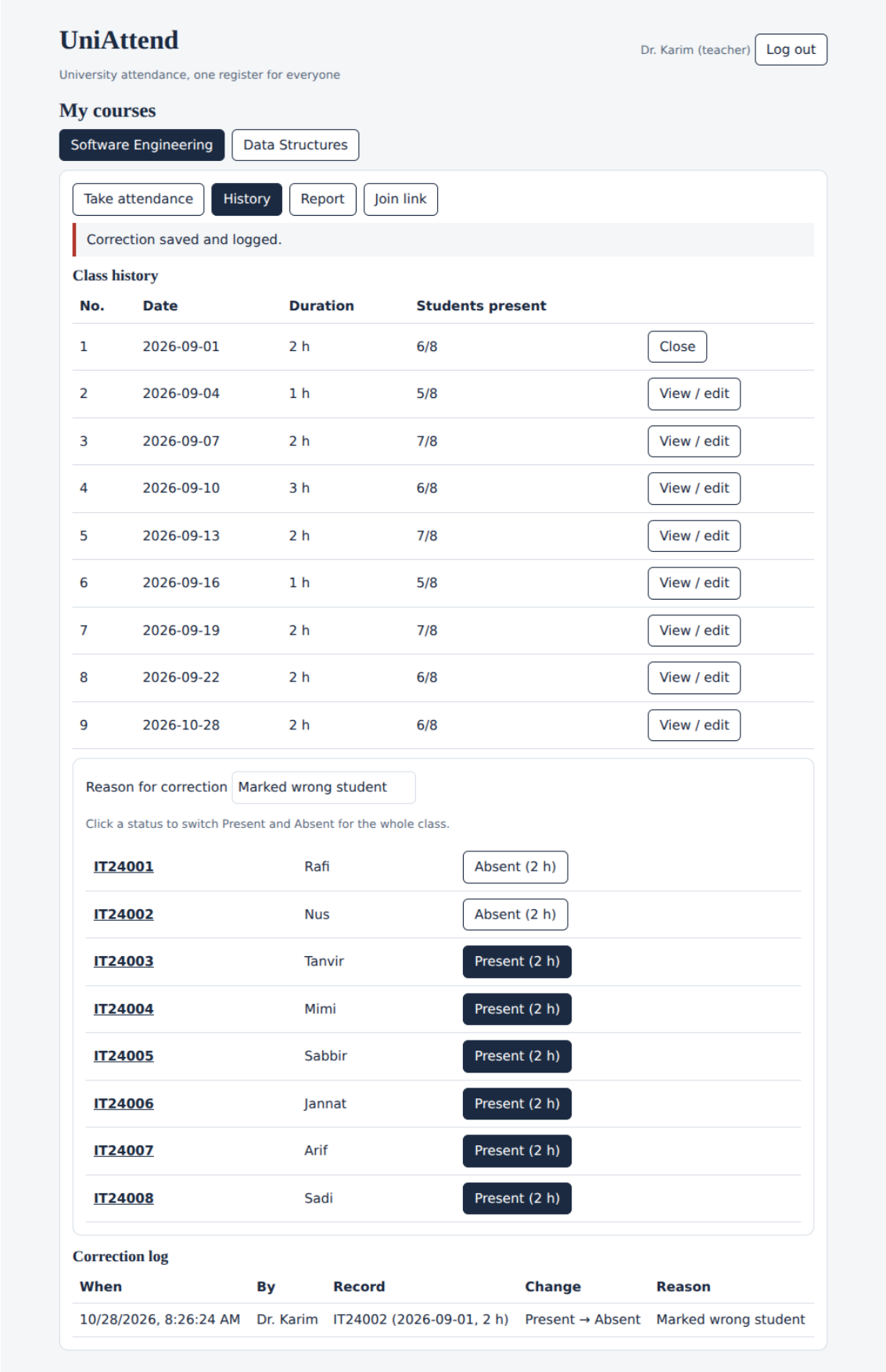


Figure 13: History, correction with reason and audit log

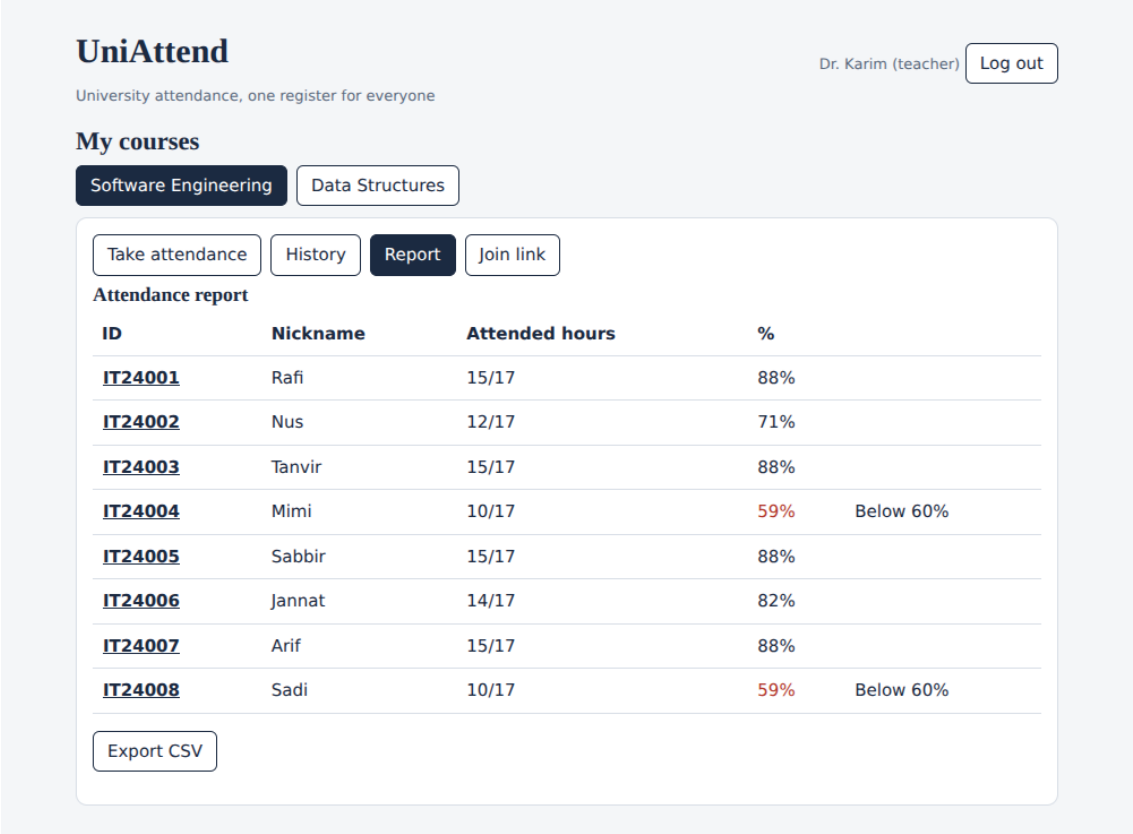


Figure 14: Teacher report in hours with low attendance flag

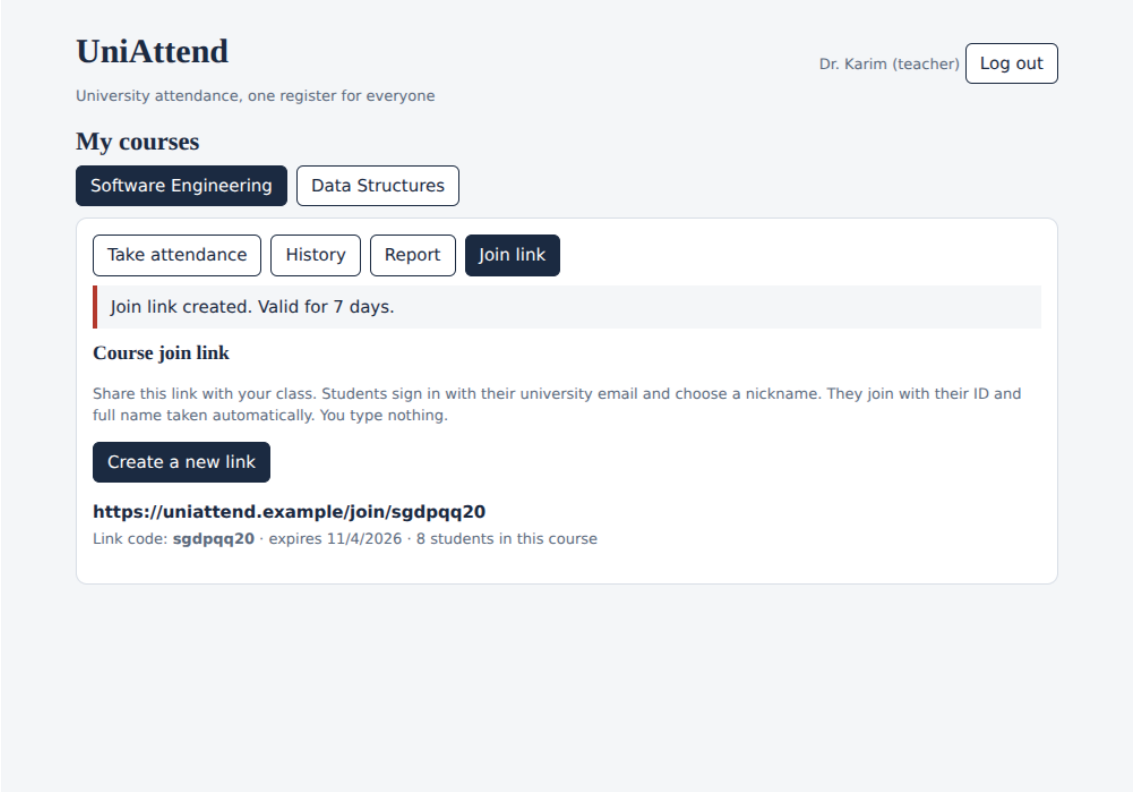


Figure 15: Teacher creates a course join link

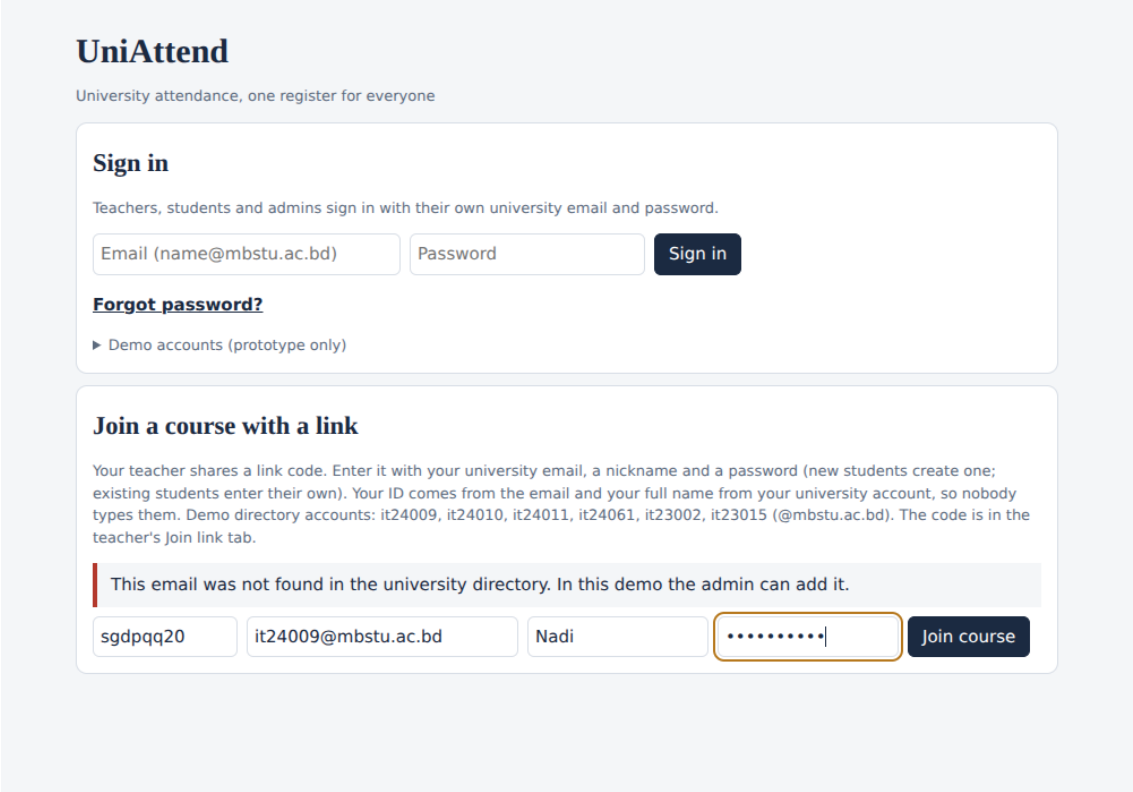


Figure 16: Student joins with university email and nickname

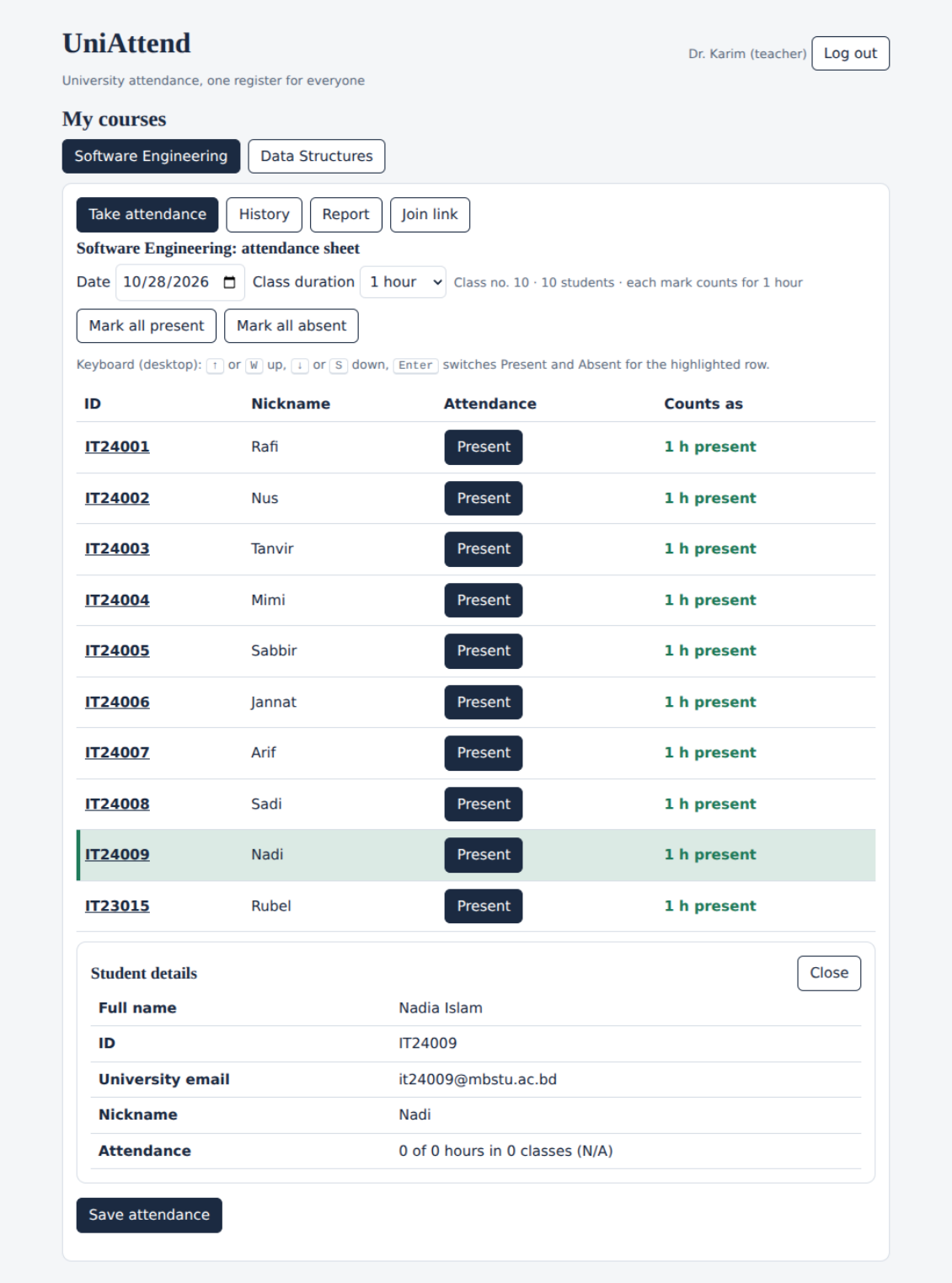


Figure 17: New student appears on the sheet with nickname and ID

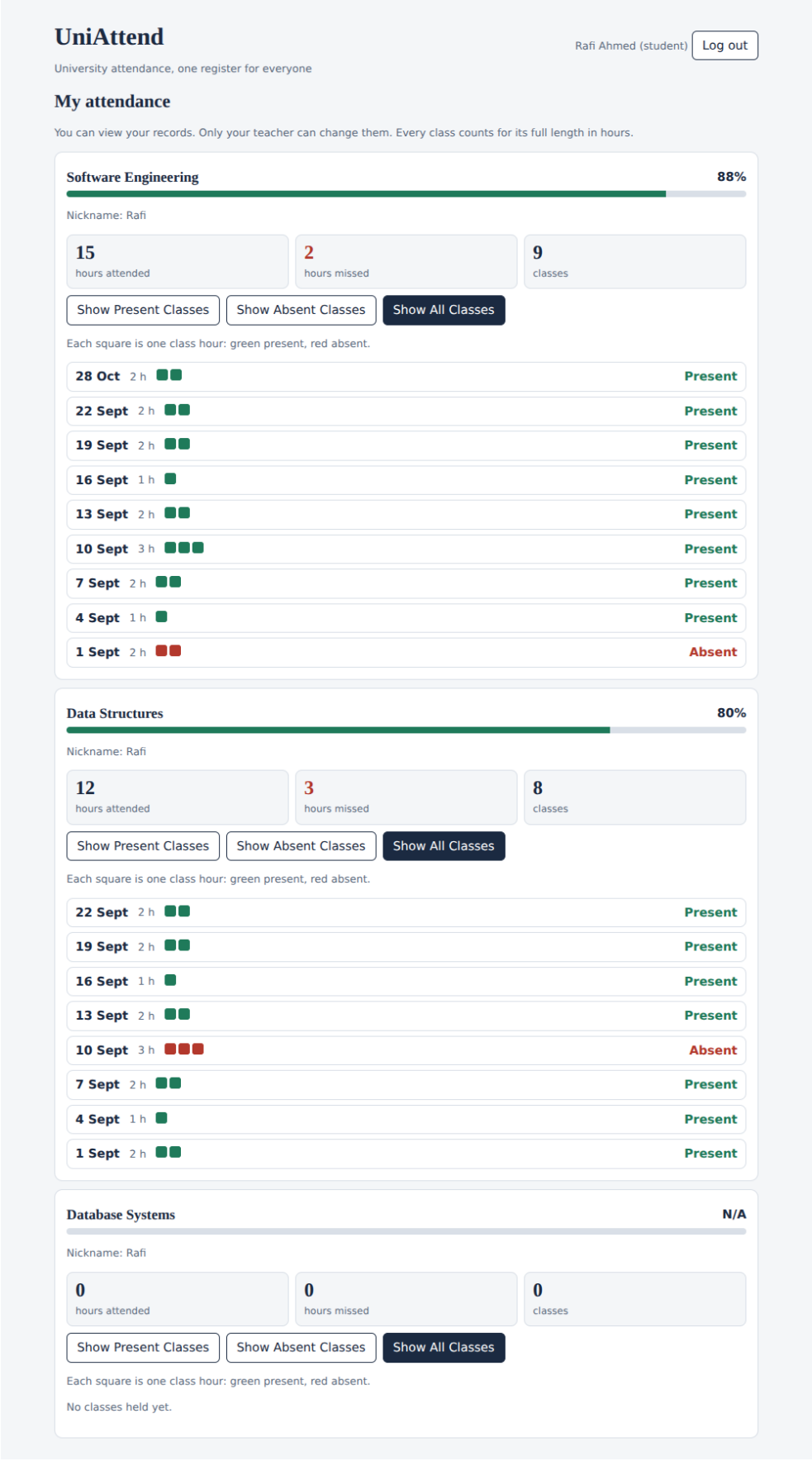


Figure 18: Student dashboard: hour tiles, class timeline with hour squares and Present / Absent / All filters

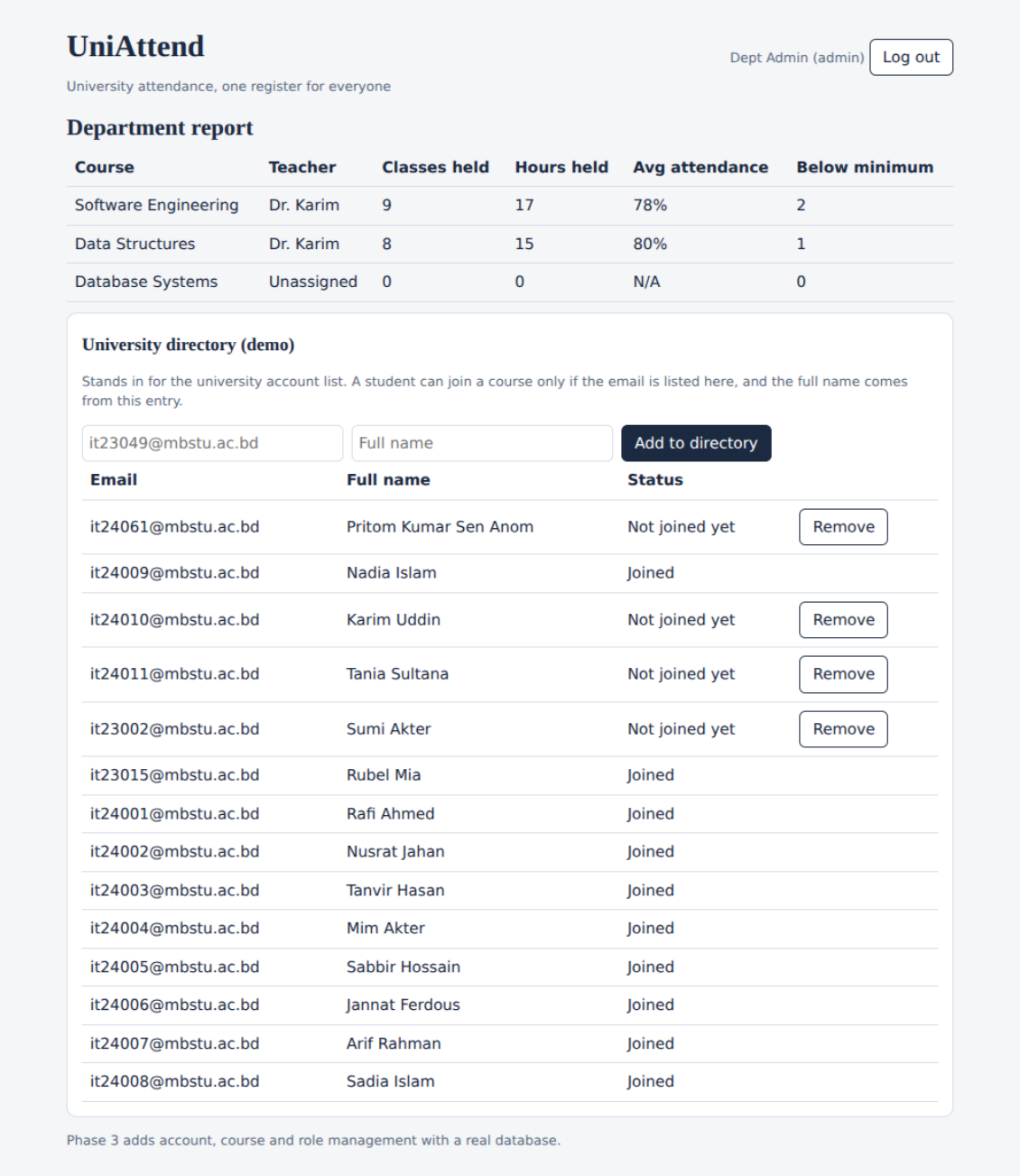


Figure 19: Admin department report and university directory

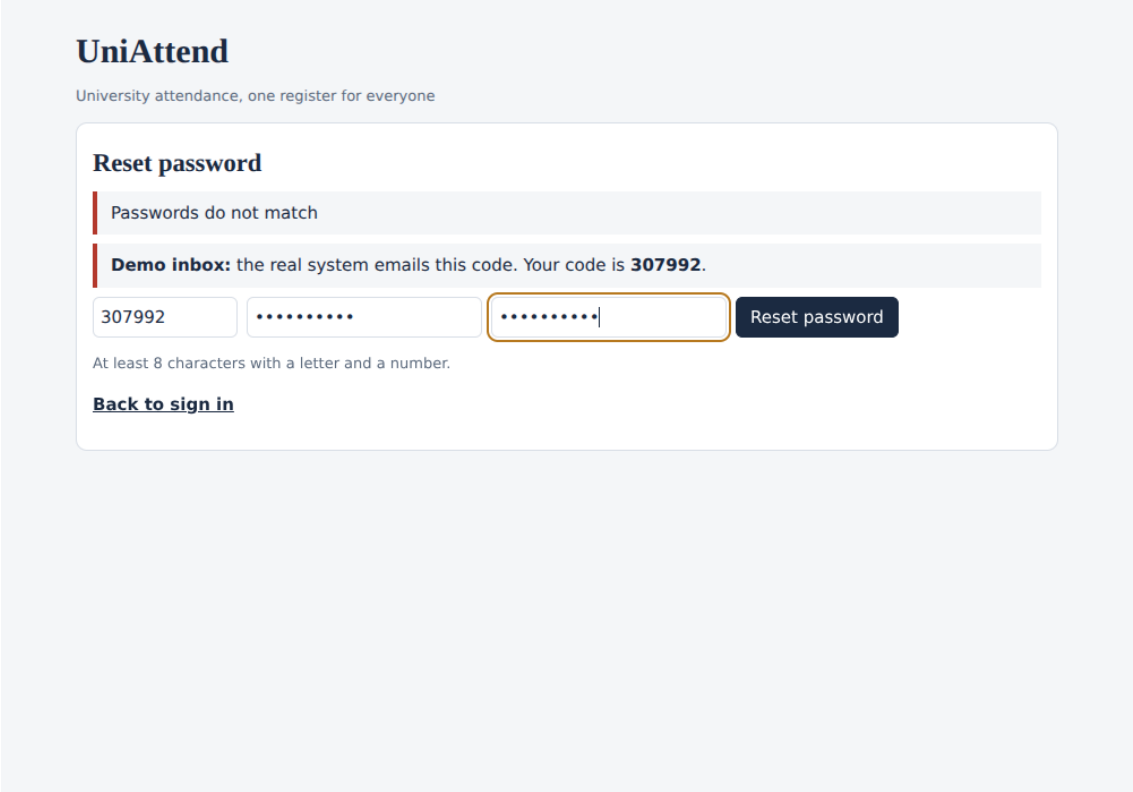


Figure 20: Forgot password: reset code and new password

13. Feasibility Study

Area	Finding
Technical	The stack (HTML, CSS, JavaScript, Java with Spring Boot, MySQL) is mature, free and well documented. The core rules (hour-weighted percentage, corrections, join link, login) are already working in the prototype and covered by 26 automated tests. The main technical risks are learning Spring Boot and Spring Security and integrating university sign-in. Both are reduced by building in small increments and by keeping email and password login as the first version.
Economic	Development tools (VS Code or IntelliJ Community, Git, GitHub, MySQL Community, Mermaid, PlantUML, LaTeX) are free. The only running cost is hosting for the backend and database: a university server or a low-cost cloud plan (assumed budget of up to BDT 2,000 per month after deployment, to be confirmed). The benefit is saved teacher time, fewer calculation errors and a record students can trust.
Operational	Teachers already take attendance every class, so the new steps are small: pick duration, mark students, save. The join link removes manual entry of student IDs and names. Students only sign in and read. Keyboard shortcuts speed up large classes on desktop.
Schedule	About 8.5 weeks are planned (20 Oct to 20 Dec 2026) with an incremental model. Version 1 is limited to attendance, history, student view, login and the join link, so the plan fits one developer.
Legal and privacy	The system stores student names, IDs, emails and attendance. Passwords are stored only as salted hashes, access is role-based, and every correction is logged. The system must follow the university's data protection and privacy rules, and exam eligibility stays a university decision.

Conclusion: the project is technically, economically, operationally and schedule-wise feasible for version 1.

14. Effort and Cost Estimate

Work item	Estimated effort (hours)
Requirements and scope	12
UML diagrams, proposal and Gantt chart	24
Interface prototype and tests	30
Database schema	12
Spring Boot API	50
Authentication and roles	24
Frontend integration	24
Testing and bug fixing	24
Documentation	16
Deployment	10
Project management and reviews	10
Total	**236**

236 hours over about 8.5 weeks is roughly 27 hours per week for one developer. As an illustration only, at an assumed rate of BDT 300 per hour the effort is worth about BDT 70,800. Tools cost nothing, and hosting is the only cash cost (see the feasibility study). These are planning estimates and will be revised after each increment.

15. Assumptions, Constraints and Dependencies

Assumptions

- Every student, teacher and admin has a university email address.
- Classes last 1 to 5 whole hours and the default minimum attendance is 60%, configurable per course.
- Users have a modern desktop or mobile browser and internet access.
- Teachers use a desktop computer for the keyboard shortcuts; phones use touch.

Constraints

- One developer, fixed deadlines, free or low-cost tools.
- The software records attendance and calculates percentages only. The university policy decides exam eligibility.
- No integration with the university ERP in version 1.

Dependencies

- Access to the university account list or sign-in (the prototype simulates it with an admin-managed directory).
- A server or cloud account to host Spring Boot and MySQL.

16. Quality Assurance and Configuration Management

- **Version control:** Git and GitHub, small commits with clear messages, one branch per increment, merged only after the tests pass.
- **Testing:** unit tests for business rules (already automated), integration tests for the API, and user acceptance testing with a teacher and students.
- **Definition of done:** a feature is complete when it meets its requirement (FR numbers), has tests, passes them, and is documented in the README.
- **Change control:** requirement changes are recorded in the proposal and traced to tests and diagrams.
- **Security checks:** hashed passwords, lockout after wrong attempts, role checks on the server, no sensitive data in the repository.
- **Backups:** regular database backup after deployment.

17. Conclusion

University_Attendance is a realistic and extendable project. Version 1 delivers the three most valuable features first, and later increments add the real backend, admin tools and reports. Future ideas include QR-based check-in, risk prediction for students near the minimum and excused-absence requests.

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